

1) not explicitly labeled

2) Kmeans and PCA

3) A centroid is the mean of all the points in a single cluster of data points

4) the stars are the initial centroids before you minimize distance. the blue circles are the dataset plotted

5) basically it finds the closest centroid to a data point so you can assign the point to that centroid

6) centroids: arrays , X: array , np.square: array, min_distance: integer , idx: array

7) makes sure the previous function on line 8 works

8) the plot after line 10 has all the data assigned to there nearest centroids

9) it takes every point assigned to a cluster k and finds the mean of those points to find the center of the cluster.

10) runs the functions to find the ideal centroid points until the centroids converge to their ideal point.

11)

```
from sklearn.cluster import KMeans
```

```
km1 = KMeans(3)
```

```
km1.fit(X1)
```

12)

N_clusters: number of clusters and centroids

Init: the method to pick initial centroid locations

N_init: how many times the model will initialize centroids and rerun

Max_iter: how many iterations for a single run

Tol: essentially if the centroid isn't moving a significant amount the model will stop

Random_state: same as other random states

Algorithm: kmeans algo

13) How do you divide an entire list by an integer in line 30 and why do they do that? I don't understand the numpy methods, what do they do? How do you get the image data itself and can you use file types aside from png?

14) 2d array

15) standard normalization

16) each row in a dataset is a vector, that represents one observation with certain features in mind

17) all objects returned are arrays. The function basically does PCA. it centers the data by scaling means down to 0, finds a covariance matrix, and returns parameters and the principal components themselves.

18) the paragraph talks about reducing dimensionality so that you can plot data. We can't actually visualize data in more than 3D

19) that graph is the elbow method, it shows you how the error changes with different amounts of clusters

20) from sklearn.decomposition import PCA

```
pca = PCA(n_components = 2)
```

```
pca_result = pca.fit_transform(X3)
```

21) X3 is the normalized and pre processed version of the dataset

22) the result has 50 rows and 2 columns. The decimals 0.871 and .129 tell how much variation each principle component covers . The first one 87% and the second one 13%

23)

24)from sklearn.datasets import load_sample_image

```
from sklearn.cluster import KMeans
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
china = load_sample_image("china.jpg")
```

```
plt.imshow(china)
```

```
X = china.reshape(-1, 3).astype(np.float64)
```

```
K = 64
```

```
kmeans = KMeans(n_clusters=K, random_state=42, n_init='auto')
```

```
kmeans.fit(X)
```

```
compressed = kmeans.cluster_centers_[kmeans.labels_]
```

```
compressed_img = compressed.reshape(china.shape).astype(np.uint8)
```

```
plt.imshow(compressed_img)
```

25) no time :(